

Yuxin Li

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EDUCATION

University of Pennsylvania

Master of Science in Engineering, Mechanical Engineering and Applied Mechanics (Robotics track) Philadelphia, PA
Sep 2023 – May 2025

Imperial College London

Data Science Summer Course Online
Jun 2022 – Aug 2022

University of Nottingham, UK

Exchange Student Nottingham, UK
Sep 2021 – Jul 2022

University of Nottingham, Ningbo (UNNC)

BEng (Hons) Mechanical, Material and Manufacturing Engineering Ningbo, China
Sep 2019 – Jul 2023

SCHOLARSHIPS

2022 Provost's scholarship (Rank Top 5%), UNNC

RELEVANT COURSES

Machine Learning, Control and Optimization with Application in Robotics, Machine Perception, Learning in Robotics, Advanced Robotics, Learning and Control for Adaptive and Reactive Robots, Design of Mechatronic Systems

RESEARCH INTERESTS

Robot manipulation; robot learning; vision-language-action models; task and motion planning; dynamical system-based motion policies; safe human-robot interaction.

PUBLICATIONS

Preprints under review

P1. **Yuxin Li**, Sunan Sun, and Nadia Figueroa. "Learned Modulation of SO(3) Dynamical System Policies for Orientational Obstacle Avoidance." *Under Review, IEEE International Conference on Robotics and Automation (ICRA)*.

RESEARCH EXPERIENCE

GRASP Lab at the University of Pennsylvania

Research Assistant (Advisor: Dr. Nadia Figueroa) Philadelphia, PA
July 2024 – Present

Routing-Correction Steering of Pretrained VLA Policies via Learned Conditioning-Space Operators

- Formulating a routing-vs-capability hypothesis for safety-relevant out-of-distribution failures in Vision-Language-Action policies, proposing an upstream steering operator $T : \mathcal{C} \rightarrow \mathcal{C}$ that maps OOD conditionings back onto the training manifold $\mathcal{M}_{\text{train}}$ before the flow-matching action head is queried, absorbing distribution shift once at the conditioning layer rather than tilting a broken downstream conditional as in score-level guidance;
- Designing a learned-critic and manifold-regularized optimization, with the danger critic B trained from demo-divergence scoring composed with VLM-as-verifier critique and the operator T parameterized compactly as an activation-space direction or $\leq 1\%$ -parameter adapter, keeping the base policy frozen and the training budget small.

Learned Modulation of SO(3) Dynamical System Policies for Orientational Obstacle Avoidance

- Developed a geometry-aware framework for modulation of orientation motion policies on the unit quaternion manifold $\mathbb{S}^3$, decomposing corrections into a compact geometric parameter space within the attractor-aligned quaternion tangent frame;
- Designed a Gaussian Process regression pipeline that learns the mapping from orientation state to the extracted modulation parameters using a geodesic kernel, formally preserving the Lyapunov stability certificates of the nominal policy;
- Derived a closed-form online orientation safety filter that pulls back a Euclidean obstacle field onto SO(3) via the rigid-body point map, combining a modulation operator with a repulsion term for dynamic obstacles;
- Validated on a 7-DoF Franka Panda across three obstacle-avoidance scenarios (pouring, cross-table traversal, book placing), comparing against no-modulation and position-only modulation baselines.

Manipulability-Guided Motion Policy Optimization

- Acquired and pre-processed trajectory data from a UMI gripper and a motion capture system, and implemented a Quadratic Programming (QP)-based Inverse Kinematics solver for precise, real-time trajectory tracking.
- Engineered a novel manipulability-gradient optimization method to algorithmically modify human-demonstrated motions for safe, singularity-avoiding execution on a 7-DOF Franka Emika robotic arm.
- Implemented a robust motion planning algorithm, utilizing the combination of Gaussian Mixture Model clustering and Linear Parameter Varying Dynamical Systems to systematically encode complex human-demonstrated movements.

Rushworth's Lab at the University of Nottingham, Ningbo

Project Coordinator (Advisor: Dr. Adam Rushworth)

Ningbo, China
Sep 2022 – Jul 2023

Design and navigation of 4-wheel drive omni-directional robotic platform

- Spearheaded the design and fabrication of a custom omni-directional mobile robot, encompassing SolidWorks mechanical design, precise component selection, and full system assembly to achieve enhanced maneuverability and payload capacity;
- Engineered a customized ROS (Robot Operating System) navigation stack, improving SLAM and localization algorithms to enable robust, fully-autonomous navigation and path planning in cluttered or dynamic environments.

Dunant's Lab at University of Nottingham, Ningbo

Research Assistant (Advisor: Dr. Dunant Halim)

Ningbo, China
Jun 2021 – Aug 2021

Active Suspension Control Architecture and Mechanical Design

- Developed high-fidelity MATLAB/Simulink models for a motorcycle shock absorber system and implemented PID control algorithms to optimize the active suspension performance for an Electric Vehicle prototype;
- Executed mechanical design via SolidWorks to create and refine motorcycle frame component CAD models, followed by FEA structural analysis to ensure optimal integration and structural integrity of the suspension system.

COURSE PROJECT

Robotic Grasp and Placement Challenge

MEAM 5200 *Introduction to Robotics* Final Project

Philadelphia, PA
Oct 2023 – Dec 2023

- Engineered a comprehensive control system in Python for the Franka Emika Panda 7-DOF robot, integrating a real-time Inverse Kinematics solver and a perception-based object detection pipeline to execute dynamic pick-and-place tasks;
- Developed robust manipulation and stacking algorithms that incorporate real-time environmental compensation via sensor fusion to handle positioning uncertainty, with validation across Gazebo simulation and physical hardware.

4WS-4WD Autonomous Racing Vehicle Control

MEAM 5170 *Advanced Control and Optimization* Final Project

Philadelphia, PA
Oct 2023 – Dec 2023

- Constructed a high-fidelity nonlinear vehicle dynamics model for a 4-wheel steering / 4-wheel drive (4WS-4WD) autonomous racing platform using the PyDrake simulation and control framework;
- Implemented a stable, minimum-time Model Predictive Control system incorporating track constraints, utilizing the SNOPT solver to efficiently compute control inputs.

COMPETITION EXPERIENCE

Formula Student Electric China

Team Member, 2021 Season Third Prize Winner

Ningbo, China
May 2021 – Aug 2021

- Engineered and implemented a distributed drive Electronic Differential System (EDS) to optimize torque vectoring and vehicle stability;
- Validated the EDS performance using MATLAB/Simulink models and integrated low-level motor control via PID control and direct PWM programming for precise actuator response;
- Integrated a LiDAR-based localization and mapping pipeline within ROS to provide the vehicle with real-time, high-precision environmental perception.

SKILLS

Programming Capabilities: Python, ROS, ROS2, MATLAB, C++

Simulation Environments: PyBullet, Gazebo, Isaac Gym, Mujoco, PyDrake

Robot Hardware: Franka Emika Panda Robot(7-DoF), UMI gripper, omnidirectional mobile base

Deep Learning Frameworks: PyTorch, TensorFlow

Others: 3D modeling and printing, SolidWorks, FEA, RViz, Blender